

Kits That Maketh Robots

Robots are making their presence felt. These indomitable marvels are not just confined to high-tech labs but being created in our own drawing rooms—thanks to robotic kits

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Remember your first toy kit with which you built a motor-run fan or lit a bulb using pencil cell batteries? Robotic kits are advanced versions of those toy kits. Whether you are a beginner, advanced user or a hobbyist, if you wish to assemble your own robot from a kit there are certain things to keep in mind.

What does a robotic kit consist of?

Robot kits contain all the components required to assemble your own bot. These include electronic components, servo motors, gears, battery, sensors, microcontrollers, mechanical frame or chassis of the robot with plastic and metal parts, etc.

“A standard kit comes with a programmable platform that is the logic processor of the bot, motors and actuators to move the bot, sensors to gather information about surroundings, power supply and PC interface kit to enable programming of the bot via PC. Other specialised kits contain these components in variation,” says Adwait Deshpande, manager-R&D, ThinkLABS.

Some robots are wired. These can be connected to a desktop PC or laptop via USB or serial cable for operation using a set of commands. Others are wireless, operated with a remote or Bluetooth.

Kits can be for educational purposes or for use in industrial applications. The foremost thing is to define the application of the kit. A beginner should obviously start with the basics and not go for an advanced kit.

Deshpande highlights, “Basic kits are closed boxes with focus on structure building and logic building. Here, the aim is to explore the project building part and make the other technical aspects as sim-

iPitara

(From ThinkLabs)

Features

- Robot development kit for school kids
- Robots made using i-Pitara kit are programmed using CIMPLETEM visual programming language software, which eliminates the need of textual coding and memorising language syntax

Price: Not mentioned



Cyton Veta 14DOF-2G

(From EnergidTechnologies)

Features

- Single bi-handed interactive humanoid manipulator with gripper
- 32-channel servo controller with 14 free channels for adding end effectors and other peripherals
- High-quality permanent magnet direct-current servo motors with integral gear reduction

Price: ₹ 400,000



Remote Operated Land Rover

(From Kits'n'Spares)

Features

- 30-rpm geared 6V DC motor coupled to the left front wheel and another coupled to the right front wheel
- Holtek encoder-decoder pair of HT12E and HT12D employing RF as well as IR principles for remote control

Price: ₹ 3000



ple as possible. Advanced kits are usually left open to hack, modify or improvise the circuit. Open access to the kit helps build add-ons and explore possibilities.”

While the basic components of many robotic kits are the same, these differ from basic- to advanced-level kits. “The primary difference between basic and advanced kits is mainly in the features available, fidelity and number of sensors, adaptability for different applications, components used and available documentation,” says Sachitanand Malewar, research scholar

at TI DSP Lab, IIT Mumbai, and director (technology), Nex Robotics.

Sudhanshu Sharma, CEO, Techntronics Education, explains, “Basic kits contain building elements usually targeted at a particular design, e.g., an arm. In these kits, the mechanical elements cannot be used to create any design other than the one specified. Programming environments in basic kits are also very simple and cannot be used to create complex programs.”

“In advanced robotic kits, the mechanical elements are generic and can be configured to create any mechanical de-